

# Krzysztof Kotlarz

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## # Summary

I am a bioinformatics scientist and biostatistician with a PhD in biological sciences, working across genomics, genetics data analysis, statistics, and machine learning. My work spans high-throughput sequencing workflows, rare disease and clinical variant interpretation, multi-omics integration, population-scale genetic analyses, and explainable AI approaches for biological discovery.

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## # Experience

**Bioinformatician** · *Ardigen* · Kraków, Poland

Jan 2025 – Present

- **Data Preparation & Multi-Omics Ingestion:** Architected robust high-dimensional multi-omics data ingestion and curation pipelines for genomic foundation models, enforcing strict metadata standards aligned with FAIR principles.
- **High-Throughput Variant Calling & Screen Optimisation:** Built high-throughput variant calling pipelines for rapid clinical and rare disease target discovery, alongside developing pipelines for precision siRNA off-target detection.
- **Nextflow Development:** Developed standardised, modular Nextflow pipelines to automate complex genomic and transcriptomic workflows, driving reproducibility and scalability across cross-functional drug discovery teams.
- **Reporting:** Built dynamic R Markdown templates to deliver decision-ready insights, integrating complex eQTL and regulatory data into concise technical reports.

**Research Assistant** · *Wrocław University of Environmental and Life Sciences* · Wrocław, Poland

Dec 2023 – Dec 2024

- **Teaching:** High-performance computing, Nextflow pipelines, Exploratory Data Analysis.
- **Projects:** Microbiome/metatranscriptome analysis.

**Biostatistician** · *Poznań University of Medical Sciences* · Poznań, Poland

Nov 2023 – Dec 2024

- **Variant Evaluation:** Execute WGS analysis for rare diseases, focusing on variant calling and interpretation to support molecular diagnostics.
- **Microarray Analysis:** Perform clinical and microarray data analysis to identify biomarkers and genetic patterns across patient cohorts.

**Biostatistician (internship)** · *Princess Margaret Cancer Centre* · Toronto, Canada

May 2023 – Oct 2024

- **Cancer Genomics:** Study of clinical biomarkers related to cancer patient outcomes, using biostatistics and machine learning algorithms.

**Data Scientist** · *Ryvü Therapeutics* · Wrocław, Poland

May 2023 – Oct 2024

- **Drug Discovery:** Statistical model design using high-performance methods related to drug discovery.
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## # Skills

**Languages & Compute:** Python, R, Nextflow, Bash

**AI & Bio-AI:** PyTorch, LLMOps, Agentic AI, Genomic Foundation Models

**Omics & Analysis:** WGS, Variant Calling, RNA-seq, GWAS, Illumina DRAGEN

**Infra & Reproducibility:** HPC, Docker, FAIR Data, Quarto

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## # Education

**PhD degree, Biological Sciences** · *Wroclaw University of Environmental and Life Sciences* · Wroclaw, Poland  
2020 – 2024

- The application of deep learning methods in the analysis of livestock genomes.

**Master's degree, Bioinformatics: Biostatistics and bioinformatics programming** · *Wroclaw University of Environmental and Life Sciences* · Wroclaw, Poland  
2020 – 2022

- The comparison of high-performance methods in the variance components estimation process of mixed linear models.

**Bachelor's degree, Bioinformatics** · *Wroclaw University of Environmental and Life Sciences* · Wroclaw, Poland  
2017 – 2020

- The influence of selected factors on mortality among patients with acute heart failure.
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## # Projects

**High-Sensitivity HPV Circulating Tumor DNA (ctDNA) Outperforms Conventional Imaging for Immediate Post-Treatment Risk Stratification in Oropharyngeal Cancer**

*International Journal of Radiation OncologyBiologyPhysics* (2026) · [link](#)

- Evaluating ctDNA accuracy against conventional imaging for immediate post-treatment risk stratification.

**Feature Selection Strategies for Deep Learning-Based Classification in Ultra-High-Dimensional Genomic Data**

*International Journal of Molecular Sciences* (2025) · [link](#)

- Implementing and evaluating feature selection methodologies for deep learning models operating on high-dimensional genomic datasets.

**Are head and neck versus abdominal paragangliomas driven by different single nucleotide events?**

*Journal of Applied Genetics* (2025) · [link](#)

- Investigating distinct single nucleotide variant (SNV) profiles driving head/neck vs. abdominal paragangliomas.

**Exploring the impact of sequence context on errors in SNP genotype calling with Whole Genome Sequence data using AI-based autoencoder approach**

*NAR Genomics and Bioinformatics* (2024) · [link](#)

- Investigating the impact of sequence context on SNP genotype calling errors in Whole Genome Sequencing data using an AI-based autoencoder approach to improve accuracy.

**Nextflow vs. plain Bash: Different Approaches to the Parallelisation of SNP Calling from the Whole Genome Sequence Data**

*NAR Genomics and Bioinformatics* (2024) · [link](#)

- Comparing Nextflow and plain Bash approaches for parallelising SNP calling from whole-genome sequence data to evaluate their performance and efficiency.

**An Explainable Deep Learning Classifier of Bovine Mastitis Based on Whole-Genome Sequence Data—Circumventing the p >> n Problem**

*International Journal of Molecular Sciences* (2024) · [link](#)

- Designing an explainable deep learning classifier to identify bovine mastitis from whole-genome sequence data, effectively circumventing the "p >> n" problem.

#### **IVIg Use Associated with Ten-Fold Reduction of Serious Infections in Multiple Myeloma Patients Treated with Anti-BCMA Bispecific Antibodies**

*Blood Cancer Discovery* (2023) · [link](#)

- Analysis based on Self-Controlled Case Series (SCCS) model in patients with multiple myeloma using BCMA BiAb antibody treatment.

#### **Better safe than sorry—Whole-genome sequencing indicates that missense variants are significant in susceptibility to COVID-19**

*PLoS ONE* (2023) · [link](#)

- Evaluating functional missense variant burdens across whole-genome sequencing cohorts to assess COVID-19 susceptibility.

#### **Application of mixed linear models for the estimation of functional effects on bovine stature based on SNP summary statistics from a whole-genome association study**

*Genetics Selection Evolution* (2022) · [link](#)

- Comparison of high-performance methods for estimating variance components of linear mixed models.

#### **Gene Variants Related to Cardiovascular and Pulmonary Diseases May Correlate with Severe Outcome of COVID-19**

*International Journal of Molecular Sciences* (2022) · [link](#)

- Interpreting and analysing data from GWAS based on WGS data.

#### **Beyond GWAS—could genetic differentiation within the allograft rejection pathway shape natural immunity to COVID-19?**

*International Journal of Molecular Sciences* (2022) · [link](#)

- Constructing and applying a handwritten HPC algorithm to Linear Mixed Models using WGS data.

#### **Identification of functional features underlying heat stress response in Sprague-Dawley rats using mixed linear models**

*Scientific Reports* (2022) · [link](#)

- Constructing and applying a hand-written HPC algorithm in the application of Linear Mixed Models using gene expression data from rats.

#### **The application of deep learning for the classification of correct and incorrect SNP genotypes from whole-genome DNA sequencing pipelines**

*Journal of Applied Genetics* (2020) · [link](#)

- Designing and applying Deep Learning models in rare-event data using SNP data from WGS.

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#### **# Achievements**

##### **Conference Scholarship**

*World Congress on Genetics Applied to Livestock Production (WCGALP 2022)* · Jul 2022

- Awarded full travel and registration coverage for presentation.

##### **Minister of Education and Science's Scholarship**

*Ministry of Education and Science, Poland* · Oct 2021

- National award for outstanding scientific achievements.

## **PRELUDIUM BIS 1 Research Grant**

*National Science Centre (NCN) · Oct 2020 – Sep 2024*

- The application of deep learning methods in the analysis of livestock genomes\* (~532,800 PLN / €125,000).  
Principal Investigator / PhD Fellow for deep learning-driven genomic frameworks.

## **Chairman, Student Bioinformatic Scientific Group (SKN Bioinformatyków)**

*Wrocław University of Environmental and Life Sciences · Oct 2021 – Jun 2022*

- Led academic initiatives, workshops, and scientific operations for the student bioinformatics association.
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## **# Conferences**

### **International Conference of Quantitative Genetics (ICQG 2024)**

- *Vienna, Austria · Poster Presentation · Jul 2024*

### **European Society of Human Genetics (ESHG 2024)**

- *Berlin, Germany · Poster Presentation · Jun 2024*

### **European Federation of Animal Science (EAAP 2023)**

- *Lyon, France · Poster Presentation · Sep 2023*

### **European Federation of Animal Science (EAAP 2022)**

- *Porto, Portugal · Oral Presentation · Sep 2022*

### **World Congress on Genetics Applied to Livestock Production (WCGALP 2022)**

- *Rotterdam, Netherlands · Oral Presentation · Jul 2022*

### **European Federation of Animal Science (EAAP 2021)**

- *Davos, Switzerland · Oral Presentation · Aug 2021*

### **European Federation of Animal Science (EAAP 2020)**

- *Online · Oral Presentation · Nov 2020*

### **Polish Bioinformatics Society (PTBI 2020)**

- *Online · Oral Presentation · Nov 2020*
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## **# Courses**

- **LARGE LANGUAGE MODELS ON SUPERCOMPUTERS** · VSC - Vienna Scientific Cluster · Sep 2024
- **MPI FOR PYTHON** · VSC - Vienna Scientific Cluster · Sep 2024
- **PYTHON FOR HPC** · VSC - Vienna Scientific Cluster · May 2024
- **FROM MACHINE LEARNING TO DEEP LEARNING: A CONCISE INTRODUCTION** · High-Performance Computing Center Stuttgart · Jun 2023
- **GOOGLE CLOUD FOR DATA ANALYST** · DaftAcademy · Dec 2022
- **KRZYSZTOF MĘDRELA WORKSHOPS PYTHON & MACHINE LEARNING** · Krzysztof Mędreła · Nov 2020